PRODUCT REQUIREMENT DOCUMENT

AEO Optimizer

AI Engine Optimization for Hospitality — UMHackathon 2026

Domain: AI Systems & Agentic Workflow Automation

Version 3.0 | April 2026

1. Project Overview

1.1 Problem Statement

Businesses today rely on human-centric optimization strategies such as SEO, review management, and marketing copy. However, AI agents are increasingly making decisions on behalf of users — for trip planning, hotel selection, and service discovery. These agents require structured, clear, and machine-readable signals to make decisions. Most hotels are still optimized for human readers, not AI decision-makers, resulting in a critical visibility gap where high-quality hotels are overlooked simply because AI systems cannot effectively interpret their data.

1.2 Target Domain

AI Systems & Agentic Workflow Automation — applied to hospitality (hotel discovery and AI-agent selection).

1.3 Proposed Solution

AEO Optimizer is a multi-agent AI system that simulates AI decision-making, evaluates a hotel's likelihood of being selected by an AI agent, and automatically generates optimized, AI-friendly content to measurably improve its visibility. The system transforms fragmented, manual hotel profiling into an intelligent, automated, and auditable optimization pipeline.

2. Background & Business Objective

2.1 Background

Traditional discovery and optimization workflows in the hospitality industry are fragmented and heavily manual. Marketing teams rely on multiple platforms — booking sites, customer reviews, internal tools — to evaluate and improve hotel listings. This process requires continuous human

interpretation of unstructured data, repeated decision-making, and coordination across disconnected systems.

With the rise of AI agents handling tasks such as trip planning, decision-making is shifting toward automated systems that depend heavily on structured, interpretable, and clearly defined signals. Most businesses are still optimized for human readers rather than AI decision-makers, creating a systemic gap.

2.2 Strategic Fit

This solution is purpose-built for Domain 1. Z.AI's GLM acts as the central reasoning engine orchestrating a multi-step agentic process: data extraction, AI-based evaluation, gap analysis, automated optimization, and validation. If GLM is removed, the system loses its ability to coordinate and execute these processes — demonstrating its critical and non-substitutable role.

3. Market Research & Industry Validation

This section presents evidence from credible third-party sources validating the market opportunity, urgency, and commercial relevance of AEO Optimizer. All statistics are sourced from industry analysts, consulting firms, and peer-reviewed market research institutions.

3.1 The AI Travel Market Is Exploding

The scale of AI adoption in travel provides the commercial context for AEO Optimizer's problem statement.

Statistic	Figure	Source
AI in Tourism market value (2024)	USD 2.95 billion	MarketsandMarkets, Dec 2024
AI in Tourism projected value (2030)	USD 13.38 billion	MarketsandMarkets, Dec 2024
CAGR of AI in Tourism (2024-2030)	28.7%	MarketsandMarkets, Dec 2024
Global travellers using AI for planning	40% of travellers worldwide	Statista / Phocuswright, 2025
VC funding in AI travel startups (H1 2025)	45% of all travel VC funding	McKinsey / Skift, Sept 2025
AI mentions in travel company annual reports	35% of Skift Travel 200 firms (up from 4% in 2022)	McKinsey, Sept 2025

Millennials & Gen Z more comfortable with AI travel	48-62% more comfortable YoY	Phocuswright, 2025
Consumers open to AI travel planning globally	Over 60%	Phocuswire / Accenture, 2024

What This Means for AEO Optimizer

- The AI travel sector is growing at nearly 3x the pace of the broader travel tech market.
- Nearly half of all new travel VC funding is flowing into AI-enabled offerings, validating investor appetite for this space.
- With 40% of travellers already using AI tools for planning decisions, the shift from human-led to agent-led hotel discovery is not hypothetical — it is already happening.

3.2 AI Agents Are Reshaping Hotel Discovery & Selection

The emergence of agentic AI is directly disrupting how hotels get discovered and selected — the core problem AEO Optimizer addresses.

- Nearly 25% of consumers are now comfortable letting an AI agent plan and book their entire trip, a figure growing rapidly as trust in AI increases. (PYMNTS Intelligence, Dec 2025)
- 36% of US adults would delegate trip planning entirely to an AI travel agent. (Forrester Research, cited in SaM Solutions, Jan 2026)
- Travel commerce is actively shifting away from search-led workflows toward agent-led execution, where AI agents not only recommend hotels but execute bookings autonomously. (PYMNTS, Dec 2025)
- The world's largest hotel chains — Marriott, Hilton, Hyatt, and Wyndham — are explicitly repositioning their direct booking strategies around AI agent compatibility, citing OTA commissions of 15-25% as the key pain point. (Financial Times / PYMNTS, Dec 2025)
- B2B tools helping travel brands achieve visibility on AI platforms are emerging as a distinct, fast-growing market category. (Phocuswire, Nov 2025)

3.3 Hotels Face a Structural AI Visibility Gap

The specific problem AEO Optimizer solves — hotels being invisible to AI agents due to poor content structure — is documented and growing in urgency across the hospitality industry.

- Hotels that fail to adapt risk being invisible at the very stage when guests are making decisions. Properties whose content cannot be easily discovered, understood, and trusted by AI systems simply will not be shown. (HiJiffy, Oct 2025)
- When AI agents encounter information gaps — such as missing late checkout policies, EV charger availability, or vague amenity descriptions — they either omit the property or generate confidence-eroding generic responses. (Platform Aeronaut, Aug 2025)

- AI thrives on structured, clear, and concise information. Hotels that do not structure their content with machine-readable signals such as Schema.org markup, FAQ formats, and clear pricing context are systematically disadvantaged in AI-generated recommendations. (Inside Hospitality Solutions, Aug 2025)
- Hotels implementing AI-optimized content and concierge systems report 30% reductions in routine guest inquiries, demonstrating that AI-readable content directly improves operational outcomes. (Tredence / AHLA, 2024)

3.4 Answer Engine Optimization (AEO) Is a Validated Commercial Discipline

AEO Optimizer's core methodology — optimizing content for AI-agent selection rather than human search — is grounded in a fast-growing professional practice validated by leading marketing research institutions.

- Answer Engine Optimization is the practice of structuring content so AI-powered tools such as ChatGPT, Perplexity, Bing Copilot, and Google AI Overviews can extract and cite it as a trusted response. This is the exact capability AEO Optimizer automates for hotels. (Amsive, June 2025)
- Entity-driven, structured content has been shown to improve AI citation likelihood by more than 35% across major generative search platforms. (Search Engine Land, Oct 2025, cited in Zensciences)
- Over 65% of searches now end without a single click, as users receive answers directly from AI systems — making AEO-readiness a critical survival issue for hospitality businesses. (O8 Agency, Apr 2026)
- Over 60% of consumers use AI chatbots for research before making purchasing decisions, meaning unoptimized hotel content is actively losing bookings to better-structured competitors. (FastHippo Media, Jan 2026)
- Most businesses have not yet realized the value of AEO, creating a significant first-mover advantage for those who optimize now. (FastHippo Media, Jan 2026)

3.5 Hospitality Industry Is Actively Investing in AI Readiness

Hotels are not passive observers — industry data shows active investment in AI visibility solutions, confirming demand for AEO Optimizer's product.

- 52% of hospitality and travel marketers plan to invest in AI-driven personalization by the end of 2025. (Skift Survey, cited in Sendbird, 2025)
- Between 2018 and 2024, 65% of all tech investment in global travel and mobility was directed at AI and machine learning — the single largest share of any technology category. (Statista, 2024)
- A 2023 survey found AI was the area in which hotel chains globally expected the most innovation in the following two years — and that expectation is now being realized with active deployments across major chains. (Statista, 2024)
- Expedia, Booking.com, MakeMyTrip, and Trip.com have all deployed AI-driven discovery tools, meaning hotels that are not AI-optimized are already losing visibility on the world's largest booking platforms. (Phocuswire, Nov 2025)

3.6 Market Gap: No Automated AEO Tool Exists for Hotels

Despite clear and growing demand, the market for automated, multi-agent AEO optimization tools purpose-built for hospitality is nascent. Existing solutions are either manual consulting engagements, generic SEO tools unadapted to AI-agent signals, or OTA-specific listing optimizers that do not address the AI decision-making layer. AEO Optimizer addresses this gap directly by:

- Simulating the AI decision-making process rather than guessing at it
- Providing criterion-level gap analysis specific to hospitality AI signals
- Automating content regeneration within a human-approved agentic workflow
- Producing measurable before/after scores that justify ROI to hotel operators

Market Validation Summary

- The AI in travel market will grow 4.5x from USD 2.95B to USD 13.38B by 2030 (MarketsandMarkets, 2024).
- 40% of global travellers already use AI for planning; 25% would let AI book autonomously (Statista / PYMNTS, 2025).
- AI-agent-led hotel discovery is an active, documented commercial trend — not a future hypothesis.
- The specific problem of hotel AI-invisibility is validated by hospitality industry publications, AEO practitioners, and major hotel chains' strategic responses.
- No purpose-built, automated AEO tool for hotels currently exists at scale — representing a clear, defensible market opportunity.

4. Product Purpose & Users

4.1 Main Goal

Enable hotels to optimize their digital presence for AI-agent selection through simulation, scoring, and automated content improvement — with measurable before/after results.

4.2 Intended Users

User Type	Primary Need
Hotel Owners	Understand why their hotel is not selected by AI travel agents
Marketing Teams	Generate AI-readable optimized content at scale
Hospitality Operators	Benchmark and improve multiple properties systematically

5. User Stories & Use Cases

5.1 Primary User Stories

- As a hotel owner, I want to understand why my hotel is not selected by AI agents so that I can improve my visibility.
- As a marketing manager, I want to receive optimized content suggestions so that I can update my hotel profile without guessing.
- As an operator, I want to simulate my hotel across multiple traveller query types so I can understand my strengths and gaps by audience.

5.2 Use Case Flows

Evaluation Flow

User inputs hotel URL or data + target query → system simulates AI agent decision-making → outputs score breakdown and reasoning.

Optimization Flow

System identifies gap signals → generates improved content per criterion → user reviews via Human Approval Workflow → system re-evaluates and shows delta score.

Multi-Query Simulation Flow

User selects multiple traveller personas (e.g. budget traveller, luxury stay, family trip) → system evaluates hotel profile against each query type → outputs comparative score matrix showing which segments the hotel serves best and where gaps exist.

6. System Functionalities

6.1 Multi-Agent Workflow Engine

The system operates as a stateful multi-agent pipeline where GLM orchestrates reasoning across six sequential stages:

Agent	Name	Role	Technology
N0	Input Handler	Accepts hotel URL/data; handles missing inputs with graceful fallback	System Node
Agent 1	Data Aggregation Agent	Scrapes or ingests hotel profile data; falls back to mock data on failure	LLM + Tooling
Agent 2	AI Decision Simulator	Simulates how an AI travel agent would evaluate the hotel for a given query	LLM (GLM)
Agent 2.5	Gap Analyzer	Identifies missing or weak signals from the evaluation output	LLM
Agent 3	Optimization Agent	Rewrites and enhances hotel profile content to address identified gaps	LLM

Agent 4	Validator Agent	Checks optimized content for quality, naturalness, and confidence score	LLM
Agent 5	Re-simulation Agent	Re-runs the AI decision simulation on the optimized profile	LLM (GLM)
HITL	Human Approval Node	Allows human review and override before finalizing output	Human-in-the-Loop

6.2 Key Features

- AI Decision Simulation — simulates AI agent scoring across configurable traveller query types
- Content Optimization Engine — automatically rewrites hotel descriptions, amenity lists, and positioning copy
- Before/After Scoring System — quantifiable improvement delta displayed on dashboard
- Human Approval Workflow — human-in-the-loop override before deployment
- Multi-Query Simulation — evaluates hotel across budget, luxury, family, and business traveller personas simultaneously
- Web Dashboard Interface — step-by-step transformation display from original to optimized profile

6.3 AI Model & Prompt Design

Model Selection

Z.AI GLM is used as the core reasoning engine due to its strength in multi-step reasoning, contextual understanding, and structured output generation. GLM is non-substitutable — removing it collapses the workflow's ability to evaluate, reason, and generate structured outputs.

Prompting Strategy

Multi-step agentic prompting is used where each agent has a specialized role (evaluation, optimization, validation), ensuring modular reasoning. Each agent prompt enforces a consistent output schema (JSON) for downstream processing.

Context & Input Handling

Inputs such as reviews and descriptions are chunked and summarized before processing. Maximum input size is controlled via batching and summarization to manage token usage.

Fallback & Failure Behavior

- Retry mechanism for invalid or malformed LLM outputs
- Confidence scoring on optimization outputs — low-confidence outputs are flagged
- Human-in-the-loop override for any stage
- Graceful error messages and mock data fallback for incomplete or unscrapable input

7. Evaluation & Accuracy Framework

This section defines how the system's credibility and accuracy are measured. Since AEO Optimizer simulates AI decision-making rather than replicating a real-world algorithm, 'accuracy' is operationalized across four measurable dimensions.

7.1 Evaluation Dimensions

Dimension	What It Measures	Target Benchmark
Scoring Consistency	Variance in score when same hotel is evaluated multiple times (n=5 runs)	Std. deviation ≤ 3 points
Before/After Delta	Mean score improvement after optimization across test dataset	Average delta ≥ +15 points
Multi-Query Sensitivity	Score variance across different traveller query types for same hotel	High variance expected (≥ 20 pts spread)
Human-AI Rank Correlation	Spearman correlation between human rankers and AI scores across 5 hotels	Target rho ≥ 0.70

7.2 Test Dataset

A dataset of 10 hotel profiles with intentionally varied quality will be constructed or scraped from publicly available sources to serve as the evaluation benchmark:

Profile Type	Description	Expected Score Range
Profile A	5-star hotel, detailed amenities, strong reviews, clear pricing	75 – 90
Profile B	4-star hotel, good reviews but vague amenity descriptions	60 – 74
Profile C	Budget hostel, minimal information, mixed reviews	40 – 59
Profile D	Hotel with no price info and generic, unfocused description	20 – 39
Profile E-J	Various combinations of strength and weakness across criteria	Distributed 20–85
Optimized A-J	Same profiles after running through optimization pipeline	Each should increase ≥ 10 pts

If the system scores profiles in the expected rank order (A > B > C > D), this constitutes face validity — a credible signal that the scoring criteria are meaningful and the model is reasoning correctly.

7.3 Scoring Criteria Breakdown

Every evaluation score is decomposed into five sub-criteria to ensure transparency and internal consistency:

Criterion	Description	Weight	Example Signal
Price Clarity	Is pricing clearly stated and contextualised?	20%	RM 250/night weekday, RM 320 weekend
Location Relevance	Is proximity to landmarks and transport stated?	20%	500m from LRT, 10 min to KLCC
Amenity Match	Do listed amenities match the target query?	20%	Family: pool, kids club, cot available
Review Sentiment	Are reviews summarised with clear sentiment signals?	20%	4.3/5 — praised for cleanliness & service
Audience Alignment	Does the content clearly address the target traveller type?	20%	Positioned for business travellers

Sub-scores are summed to produce the overall 0-100 score. The sub-score breakdown is displayed on the dashboard alongside the overall score, allowing users to see exactly which criteria are driving poor or strong performance.

7.4 Consistency Testing Protocol

To validate scoring reliability, the following protocol is executed during QA:

1. Select 3 hotel profiles from the test dataset (one high, one medium, one low scoring).
2. Run each profile through the AI Decision Simulator 5 times with identical inputs.
3. Record the score from each run and compute mean and standard deviation.
4. Accept if std. deviation ≤ 3 points; flag for prompt engineering review if > 5 points.
5. Document results in the QA Testing Document with pass/fail status per profile.

7.5 Human Baseline Comparison

To provide a human-grounded credibility benchmark:

6. Three team members independently rank the same 5 hotels for a fixed query (e.g. 'best family hotel in KL under RM300/night').
7. The AI system scores and ranks the same 5 hotels.
8. Spearman rank correlation is computed between the average human ranking and the AI ranking.
9. A result of 0.70 or above is considered a strong positive result and will be reported in the pitch.

Key Accuracy Claim for Judges

- We do not claim to replicate any real booking algorithm.
- Our claim: the system reliably identifies content gaps and produces measurably higher AI-readability scores after optimization.
- All claims are backed by empirical results from the 10-hotel test dataset documented in the QA Testing Document.

8. Scope Definition

8.1 Features Included

- Multi-agent workflow system (6 agents + HITL node)
- AI-based evaluation scoring with 5-criterion sub-score breakdown
- Automated content rewriting and optimization
- Before/after score delta display
- Multi-query simulation across traveller personas
- Accuracy & consistency testing framework (10-hotel dataset, consistency protocol, human baseline)
- Simple web dashboard interface
- Hybrid data ingestion (scraping + mock fallback)

8.2 Features Not Included (Out of Scope)

- Real-time OTA platform integration (e.g. Booking.com, Agoda API)
- Multi-user collaboration and team accounts
- Full website deployment automation
- Advanced competitor benchmarking (future roadmap)
- Mobile application

9. Assumptions & Constraints

Constraint	Detail & Mitigation
LLM Token Cost	Token usage optimized through summarization and query batching. Repeated evaluations cached where possible.
Scraping Limitations	Some hotel sites may block scraping. Fallback to structured mock data is fully supported and documented.
Performance / Latency	Multi-agent pipeline may take 10-30 seconds per run. Progress indicators shown on dashboard. Async processing used where possible.

Scoring Ground Truth	No single ground truth exists for AI-agent selection. Accuracy is measured through consistency, delta, and human correlation metrics as defined in Section 7.
Data Availability	At least one real hotel data source (publicly accessible listing) will be used in the demo. Remaining profiles may be mock for QA purposes.

10. Risks & Mitigation Strategies

Issue 1: Scoring Reliability

AI-generated evaluation scores may appear subjective or inconsistent, leading to concerns about trustworthiness.

Mitigation

- Predefined evaluation criteria enforced in every prompt (price clarity, location, amenity match, review sentiment, audience alignment).
- Structured JSON output schema with sub-scores required from GLM at each evaluation.
- Consistency testing protocol (Section 7.4) validates variance is within acceptable bounds before demo.
- Emphasis on relative improvement (before vs after) rather than absolute score as ground truth.

Issue 2: Abstractness of Optimizing for AI Agents

The concept may be perceived as futuristic or hard to grasp without concrete grounding.

Mitigation

- Dashboard makes AI decision-making transparent: clear inputs, visible outputs, step-by-step transformation.
- Live demo uses a real hotel listing with before/after comparison and numeric score improvement.
- Multi-query simulation concretely shows different AI agent perspectives on the same hotel.
- Section 3 of this PRD provides market evidence from McKinsey, Phocuswright, MarketsandMarkets, and PYMNTS that this problem is real and urgent.

Issue 3: Over-Optimization Risk

Optimized content could become unnatural, misleading, or keyword-stuffed.

Mitigation

- Validator Agent checks content for naturalness and flags low-confidence outputs.
- Human Approval Workflow gives the user final review before any content is deployed.

- Optimization prompts explicitly instruct GLM to preserve factual accuracy and natural tone.

Issue 4: Limited Feature Depth Perception

Single-use tools may be perceived as lacking depth for hackathon evaluation.

Mitigation

- Positioned as a 3-engine decision intelligence platform: Simulation + Evaluation + Optimization.
- Multi-query simulation across traveller personas demonstrates system robustness and adaptability.
- Comprehensive accuracy framework (Section 7) adds analytical depth beyond basic tool functionality.

11. Quality Assurance & Testing Plan

11.1 Test Categories

Test Type	What Is Tested	Pass Criteria	Owner
Unit Tests	Individual agent prompt → output for each of the 6 agents	Correct JSON schema output, no empty fields	Engineering
Integration Tests	Full pipeline run end-to-end on 3 hotel profiles	Score produced, delta calculated, optimized content generated	Engineering
Consistency Tests	5 repeated runs on 3 profiles (high/mid/low)	Std. deviation ≤ 3 points per profile	QA Lead
Delta Tests	Before/after score on 10-hotel dataset	Mean improvement ≥ +15 points	QA Lead
Human Baseline	AI ranking vs 3-person human ranking of 5 hotels	Spearman rho ≥ 0.70	Full Team
Edge Case Tests	Missing price, no reviews, empty description inputs	Graceful error handling, fallback to mock, no crashes	Engineering
Multi-Query Tests	Same hotel across 3 query types	Score variance ≥ 20 pts across personas	QA Lead

11.2 QA Deliverable

All test results will be documented in the Quality Assurance Testing Document with: test case ID, input, expected output, actual output, pass/fail status, and notes. This document forms part of the hackathon submission.

12. User Flow & Dashboard

12.1 Step-by-Step User Flow

10. User inputs hotel URL or structured data + selects target traveller query type(s).
11. System runs multi-agent workflow: Extract → Simulate → Analyze → Optimize → Validate → Re-simulate.
12. Dashboard displays full transformation:
    - Original Profile view
    - AI Evaluation breakdown (overall score + 5 sub-criteria)
    - Gap Analysis highlights
    - Optimized Profile with change annotations
    - Score Improvement delta (e.g. 41 → 67, +26 points)
    - Multi-query comparison matrix (if multiple personas selected)
13. User reviews optimized content via Human Approval Workflow.
14. User approves or rejects; approved content is marked ready for deployment.

12.2 Dashboard Sections

Section	Content
Original Profile	Raw ingested hotel data — description, amenities, price, reviews
AI Evaluation	Overall score (0-100) + 5-criterion sub-score breakdown with reasoning
Optimization Insights	Gap signals identified per criterion with recommended improvements
Optimized Profile	Rewritten content with diff highlighting vs original
Score Improvement	Visual before/after score delta with percentage improvement
Multi-Query Matrix	Score grid across traveller personas (budget, luxury, family, business)
Deployment Options	Export optimized content; mark as approved for use

13. Appendix — Judging Criteria Alignment

The following table maps AEO Optimizer's features to UMHackathon 2026 preliminary round judging criteria:

Judging Criterion	Weight	How AEO Optimizer Addresses It
Problem Definition & Purpose	5%	Clear problem statement with market evidence from McKinsey, MarketsandMarkets, PYMNTS, and Phocuswright (Section 3). AI-agent visibility gap in

		hospitality is a documented, growing problem.
Target Users & User Stories	8%	Three user types with distinct stories. Use case flows documented for Evaluation, Optimization, and Multi-Query flows.
Originality, Innovation & Value	10%	AEO (AI Engine Optimization) is a novel, emerging discipline. Multi-agent simulation of AI decision-making for hospitality is genuinely original. Market research confirms no automated equivalent exists at scale.
Feature Prioritization & MVP Scope	7%	Clear in-scope / out-of-scope delineation. MVP includes core pipeline + accuracy framework. Advanced features deferred to roadmap.
System Logic & Architecture	7%	6-agent pipeline with defined roles, data flow, and conditional logic (HITL, retry, fallback) fully documented.
System Schema & Design	6%	Structured JSON schemas enforced at every agent boundary. Sub-score schema enables auditability.
Technical Feasibility & Workflow Integration	7%	Scraping + mock fallback ensures demo reliability. GLM is central and non-substitutable. Latency managed via async processing.
Version Control & Repository Management	5%	Modular codebase organized by agent. Git repository with clear commit history and README.
Code Modularity & Structure	8%	Each agent is an independent module. Shared utilities for LLM calls, schema validation, and data ingestion.
Implementation Quality	12%	End-to-end working pipeline. Edge case handling. Confidence scoring. Human approval workflow.
Quality Assurance & Test Planning	10%	Comprehensive QA plan: 7 test categories, 10-hotel

		dataset, consistency protocol, human baseline correlation.
Pitch & Prototype Demonstration	15%	Live demo with real hotel data. Visual before/after dashboard. Concrete numeric improvements. Market research cited to validate the problem.

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